DSA0607 DATA HANDLING AND VISUALIZATION

CO3_AT2_EXPERIMENTS

Experiment 1: Outlier Detection in Real-World Data

Objective: Identify and analyze extreme values in datasets.

Task:

- **Collect a dataset such as house prices, salaries, or transaction amounts.**
- **Visualize the data using box plots and scatter plots.**
- **Apply IQR (Interquartile Range) method to detect outliers.**
- **Compare results with Z-score based outlier detection.**
- **Analyze how outliers affect mean and standard deviation.**

Aim

To identify and analyze outliers in a real-world dataset using **Box Plots** and **Scatter Plots** in Tableau and compare the results using the IQR and Z-score methods.

Procedure

1. Import the Dataset

1. Open Tableau Desktop/Public.
2. Connect to the dataset (e.g., **House_Prices.csv**).
3. Verify the fields and open **Sheet 1**.

2. Create a Box Plot

1. Drag **Category** to **Columns**.
2. Drag **Price** to **Rows**.
3. Select **Show Me → Box-and-Whisker Plot**.
4. Rename the worksheet as **Box Plot**.

3. Create a Scatter Plot

1. Open a new worksheet.
2. Drag **Area** to **Columns**.
3. Drag **Price** to **Rows**.
4. Drag **Category** to **Color**.

4. Create the Dashboard

1. Create a new dashboard.
2. Add the Box Plot and Scatter Plot.
3. Save the dashboard.

Explanation

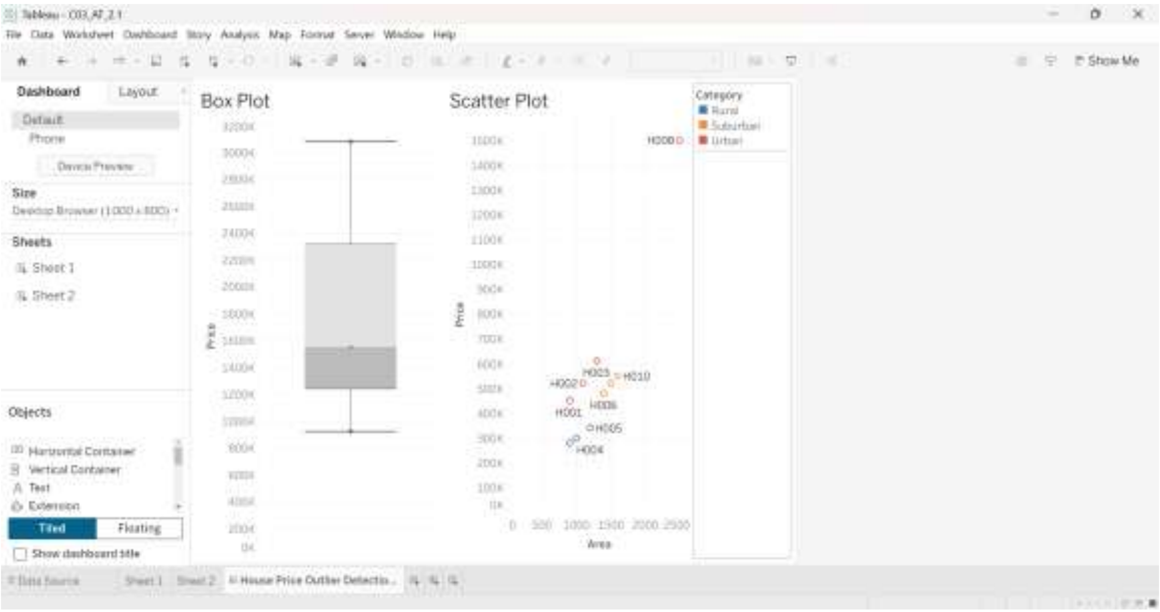
Box Plot

A box plot summarizes the distribution of house prices and highlights outliers using whiskers and individual points. It helps identify unusually high or low prices and compare variability among categories.

Scatter Plot

A scatter plot shows the relationship between house area and price. Outliers appear as points far away from the general pattern, helping detect abnormal observations.

Output



Result

The dashboard successfully identified outliers using Box Plots and Scatter Plots. The visualizations helped understand how extreme values influence the distribution and variability of the dataset.

Experiment 2: Correlation and Relationship Analysis

Objective: Understand relationships between variables in multivariate data.

Task:

● Use a dataset containing at least 4 numeric variables (e.g., car features like mileage, price, engine

power, weight).

● Compute the correlation matrix.

● Visualize using a heatmap.

● Plot scatter plots for highly correlated pairs.

● Interpret positive, negative, and weak correlations.

Aim

To analyze relationships among multiple numeric variables using a **Correlation Heatmap** and **Scatter Plots** in Tableau.

Procedure

1. Import the Dataset

1. Open Tableau.
2. Connect to **Car_Features.csv**.
3. Verify the fields.

2. Create a Heatmap

1. Drag **Variable1** to **Rows**.
2. Drag **Variable2** to **Columns**.
3. Drag **Correlation** to **Color**.
4. Select **Square** in the Marks card.

3. Create a Scatter Plot

1. Open a new worksheet.
2. Drag **Engine_Power** to **Columns**.
3. Drag **Price** to **Rows**.
4. Drag **Mileage** to **Color**.

4. Create the Dashboard

1. Create a dashboard.

2. Add the Heatmap and Scatter Plot.
3. Save the dashboard.

Explanation

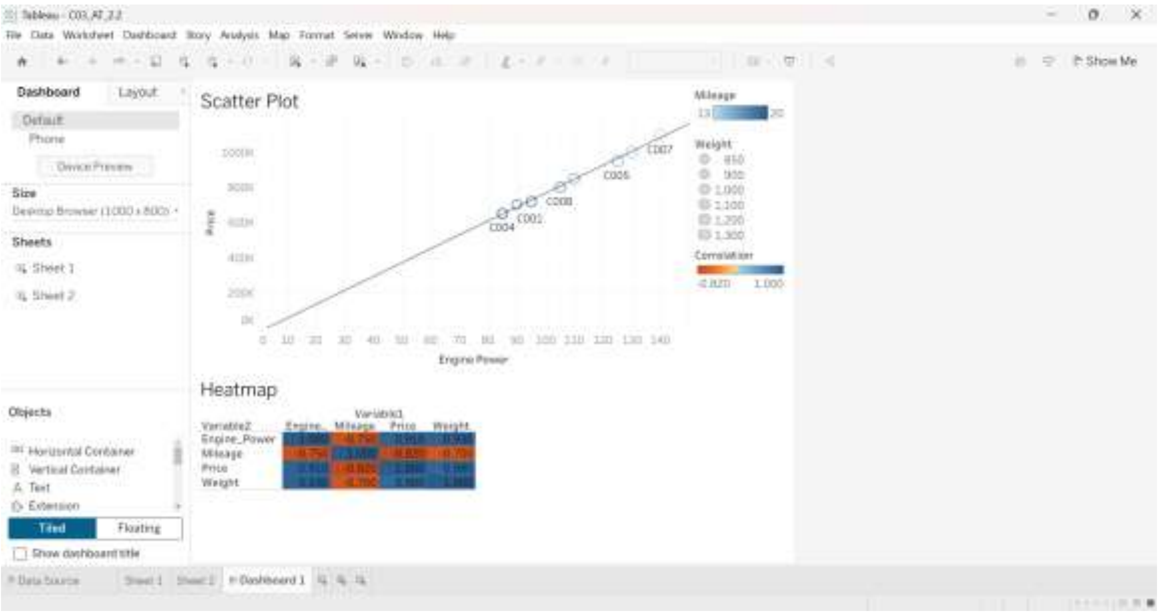
Heatmap

A heatmap displays correlation values between variables using different colors. Strong positive and negative correlations are easily identified through color intensity.

Scatter Plot

A scatter plot shows the relationship between two numeric variables. It helps identify positive, negative, or weak correlations and detect unusual observations.

Output



Result

The dashboard effectively visualizes correlations among variables and helps identify strong and weak relationships, supporting better data analysis.

Experiment 3: Time Series Trend and Seasonality Analysis

Objective: Analyze temporal patterns in sequential data.

Task:

- **Collect time series data (e.g., monthly sales, temperature, website traffic).**

- **Plot line charts to observe trends over time.**
- **Decompose the series into trend, seasonal, and residual components.**
- **Identify patterns such as seasonality or irregular fluctuations.**
- **Compare raw data vs smoothed moving average.**

Aim

To analyze time series data using **Line Charts** and **Moving Average** to identify trends and seasonal patterns.

Procedure

1. Import the Dataset

1. Open Tableau.
2. Connect to **Monthly_Sales.csv**.
3. Verify the fields.

2. Create a Line Chart

1. Drag **Month** to **Columns**.
2. Drag **Sales** to **Rows**.
3. Select **Line** in the Marks card.

3. Apply Moving Average

1. Right-click **SUM(Sales)**.
2. Select **Quick Table Calculation → Moving Average**.
3. Adjust the moving average settings.

4. Create Dashboard

1. Add the Line Chart and Moving Average chart.
2. Save the dashboard.

Explanation

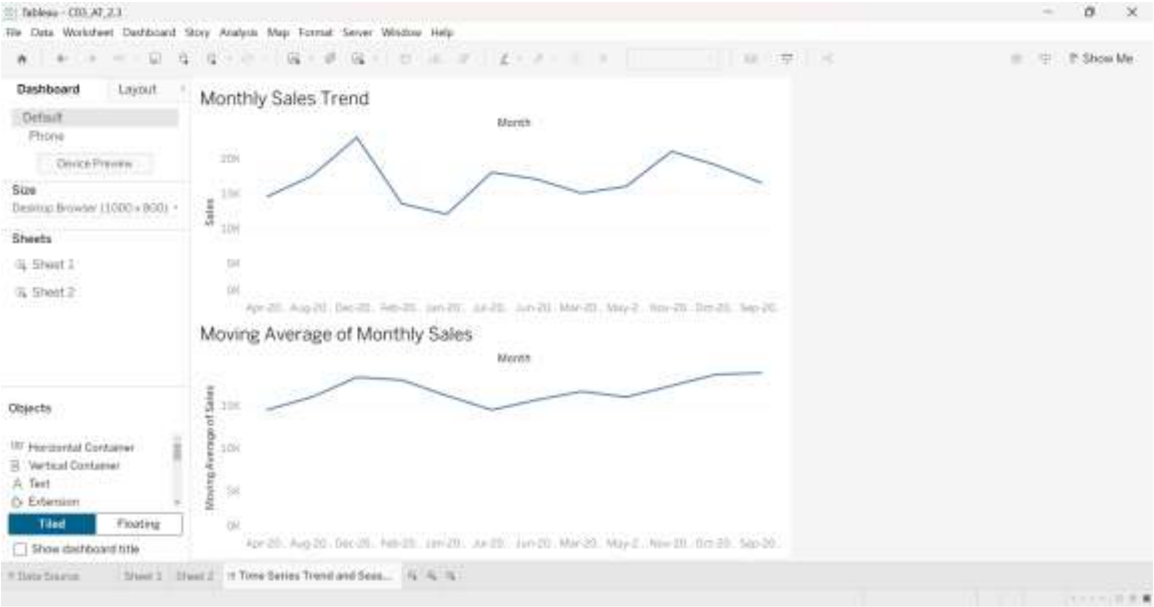
Line Chart

A line chart displays changes in sales over time, making it easy to observe increasing or decreasing trends and seasonal fluctuations.

Moving Average

A moving average smooths short-term fluctuations and highlights the overall trend. It helps compare raw data with smoothed values for better interpretation.

Output



Result

The Tableau dashboard successfully visualizes time series trends and moving averages, helping identify seasonal patterns and long-term trends.